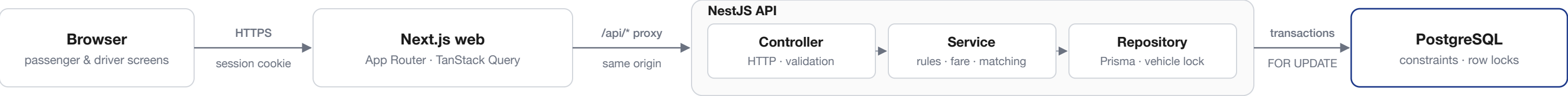
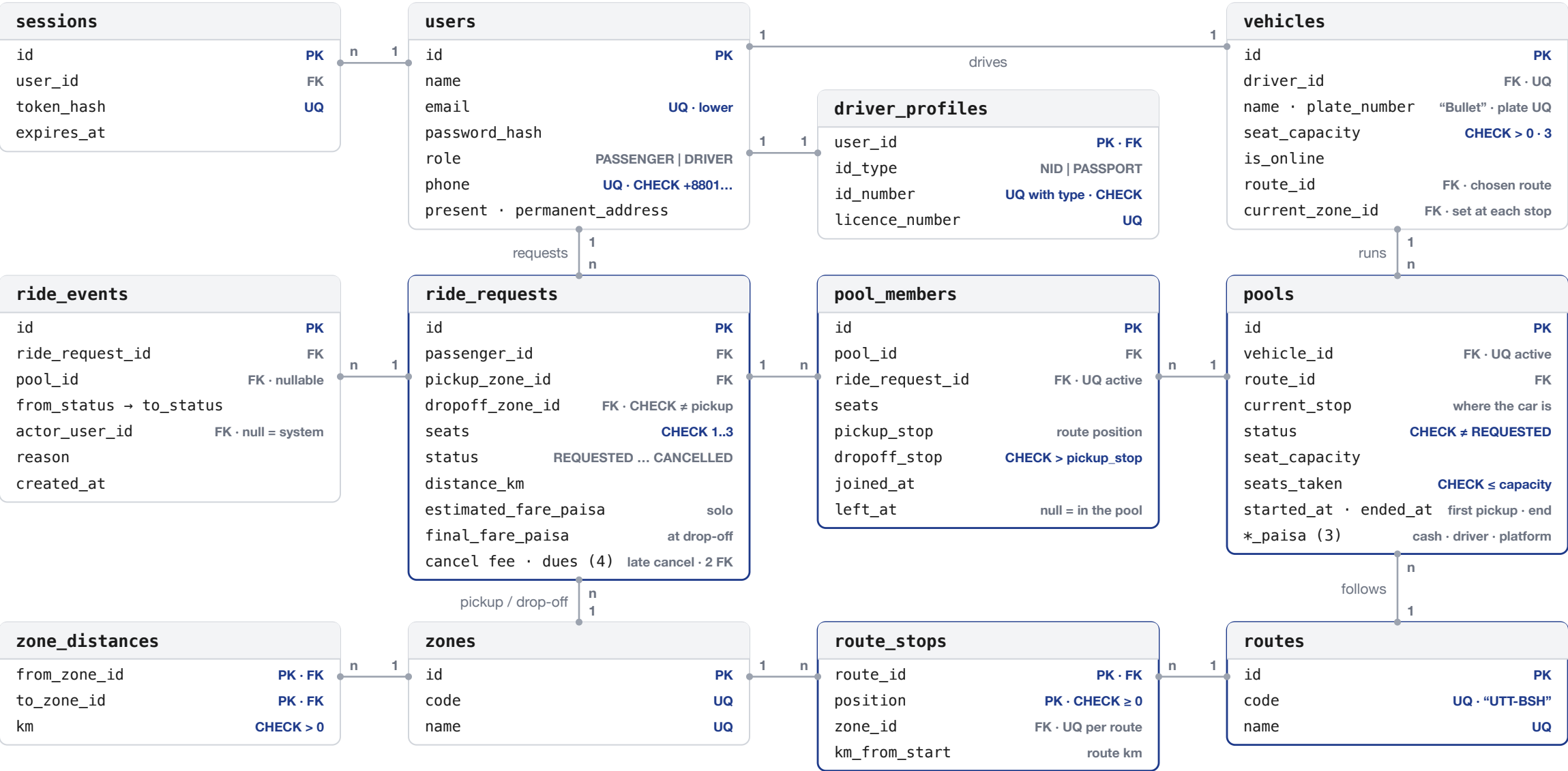


1 REQUEST PATH



Guards: session → role, then ride ownership in the service. Public: /health, /zones, /routes, /auth/signup/*, /auth/login. A lower layer never imports an upper one.

2 DATA MODEL



Integrity guarantees

- seats_taken ≤ seat_capacity (CHECK)
- pickup_stop < dropoff_stop (CHECK)
- collected = driver + platform (CHECK)
- 1 active pool per vehicle
- 1 active membership per ride
- 1 active ride per passenger
- 1 account per email · phone · ID · plate
- fee only on a cancelled ride (CHECK)
- every change: lock vehicle → check → write
- database rules hold even if code is wrong

Fare

(Tk 30 + direct km × Tk 15) × seats

–20% if you shared at least one hop

estimate = solo fare, never exceeded

driver: Tk 10/km carried + Tk 20/pickup

platform keeps the rest · never < Tk 10

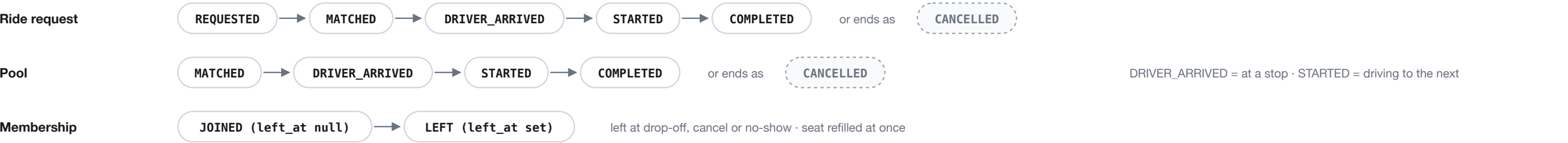
late cancel Tk 20 → the driver who came

story: Tk 240 = 180 driver + 60 platform

Legend

- PK primary · FK foreign · UQ unique
- 1 — n one-to-many
- blue border = core of pooling
- Cast: Jashim · Bullet (3 seats, UTT → BSH)
- Nusrat · Rafiq · Shirin

3 LIFECYCLES



Per passenger: DRIVER_ARRIVED = car at your stop · STARTED = on board · COMPLETED = dropped off (fare locked). Passenger cancels until picked up; driver cancels before the first pickup → riders back to REQUESTED.

Joins (R1–R4): on the route in its direction and at most +2 km / +40% round, car not past the pickup, seats free, pool active. Any other transition → 409; every passenger status change writes a ride_event.

Late cancel or no-show once the car is coming to the stop (after a 2-minute grace): Tk 20, paid with the next ride, earned by that driver. A freed seat goes straight to the next waiting rider who fits.